NETWORK SCIENCE · APRIL 2026

Stability and Collapse of Pollination Networks

A Network Science Perspective · Indian Institute of Technology Bhilai

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172 NETWORKS

6 CONTINENTS

0.960 GNN AUC

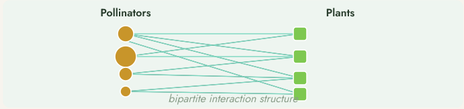
11.5K SPECIES

ABSTRACT Pollination networks sustain **~88% of flowering plants** and **75% of global food crops**. We analyse **172 real-world bipartite networks** across 6 continents using cascading co-extinction simulation, climate stress modelling, adaptive rewiring, early warning signal detection, Random Forest extinction risk prediction, **GraphSAGE-based role discovery**, and species transplantation to characterise structural vulnerability and collapse dynamics at continent scale.

1 Dataset & Network Model

Bipartite graph **G = (U ∪ V, E)**: pollinators (U), plants (V). **172 city-level datasets** from Web of Life were grouped into 6 continental meta-networks for macro-scale comparison, while all 172 were retained for network-level GNN analysis.





- **Degree Centrality** — generalist identification
- **Betweenness Centrality** — bridge species
- **Eigenvector Centrality** — systemic influence
- **NODF Nestedness** — specialist–generalist hierarchy
- **Modularity Q** — community compartmentalisation

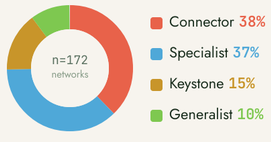
2 Continental Network Properties

Continent	Plants	Pollin.	NODF	Q	Conn.
Africa	147	737	18.2↑	0.453	0.15
Asia	543	2947	4.3	0.602	0.12
Europe	417	1988	5.4	0.627	0.13
N. America	909	1855	3.2↓	0.270↓	0.09
Oceania	127	299	9.4	0.488	0.16
S. America	1089	907	3.9	0.773↑	0.08

South America (Q=0.773) most modular; **North America** (Q=0.270) most integrated and most vulnerable to targeted attack.

3 GNN Functional Role Discovery

Two-layer **GraphSAGE** on all 172 networks. Link-prediction pre-training → k-means (k=4) → Generalist / Specialist / Connector / Keystone.



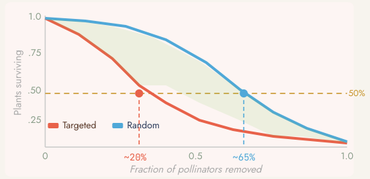
0.960 Mean GNN AUC (171/172 ≥ 0.85)

0.384 Mean Silhouette score (k=4)

Connectance–Generalism: r = 0.48, p < 0.001. Keystone prevalence is connectance-independent (r = 0.07, p = 0.345).

4 Cascading Co-extinction

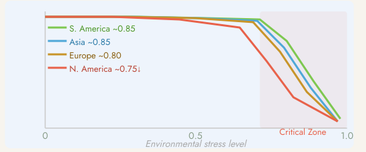
Pollinators removed sequentially; plants co-extinct when all links lost. 50 repeats for random removal average.



2.5–4× difference in network half-life. North America (Q=0.270) collapses earliest — targeted removal generates 80–300 isolated components.

5 Climate Stress Simulation

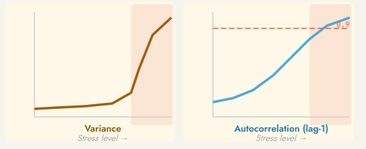
Climate stress modelled as gradual reduction in interaction strength (s: 0→1). All networks collapse rapidly beyond s ≈ 0.75–0.85.



Pollinators decline before plants across all networks — monitoring pollinator abundance provides earlier ecosystem stress warning than monitoring plant abundance.

6 Early Warning Signal Detection

Critical Slowing Down (CSD): rising variance and lag-1 autocorrelation precede collapse bifurcation. CSD autocorrelation threshold = 0.9.



- Variance rises at stress ≈ `0.70–0.75`, before critical zone at ~0.79
- Warning window: `10–30 years` before catastrophic transition
- `Oceania` shows earliest warning — limited island redundancy

7 Adaptive Rewiring (p = 0.6)

Surviving plants seek alternative pollinators proportional to degree when a partner is lost. Resilience gain = area between rewiring and no-rewiring curves.

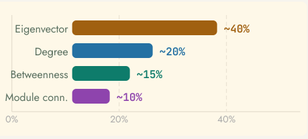


- **Largest gain:** Europe & Oceania (high nestedness)
- **Smallest:** Asia & N. America (modular, locally saturated)
- Benefit positively correlated with nestedness

Conservation: Habitat corridors enabling pollinators to encounter novel plants significantly boost resilience beyond species-level protection.

8 ML Extinction Risk Prediction

Random Forest classifier (100 trees, 5-fold CV) on structural node features. Consistent feature importance across all 6 continental networks.



Eigenvector dominates (~40%). Conserving pollinators by interaction frequency alone misses systemically critical species at strategic positions.

9 Species Transplantation

Introduced *"Unidentified sp16 M_PL_050"* into M_PL_050 target community (17 plants × 51 pollinators).



Metric	Before	After	Δ
Interactions	129	132	+3
NODF	38.94	38.06	−0.88
Modularity Q	0.4627	0.4502	−0.013
LCC Size	68	69	+1

Key constraint: transplanted species must target bridge plants (high betweenness) — not over-connected nodes — to improve resilience.

KEY FINDINGS & CONCLUSIONS

01 Continental Architecture Diversity

South America (Q=0.773) most stress-resistant. **North America** (Q=0.270) most vulnerable — 2.5–4× faster collapse under targeted vs. random removal. Africa's high nestedness (NODF=18.2) drives synchronous tipping-point collapse.

02 GNN Corpus & Eigenvector Dominance

Across 172 networks, **Connector (38%)** and **Specialist (37%)** are co-dominant regimes. **Eigenvector centrality** consistently outperforms raw degree as extinction risk predictor (~40% RF importance), arguing for eigenvector-based conservation prioritisation.

03 Early Warning & Rewiring Buffer

CSD signals detectable **10–30 years** before catastrophic collapse. **Adaptive rewiring** improves resilience across all networks — highest in nested communities. **Managed transplantation** modestly improves resilience when species bridges structural gaps.

REFERENCE

[1] J. Bascompte, P. Jordano, C. J. Melian, and J. M. Olesen, "The nested assembly of plant-animal mutualistic networks," *Proc. Natl. Acad. Sci. U.S.A.*, vol. 100, no. 16, pp. 9383-9387, 2003.

Data Source: Web of Life (www.web-of-life.es)

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Bhilai, Chhattisgarh · April 2026